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Competition scheduling patterns of emerging elite players in professional men's tennis

Thomas Perri^{a,b}, Rob Duffield^b, Alistair Murphy^b, Tom Mabon^b and Machar Reid^b

^aSchool of Sport, Exercise and Rehabilitation, Faculty of Health, University of Technology Sydney, Moore Park, Sydney, NSW, Australia; ^bSports Science and Sports Medicine Unit, Tennis Australia, Melbourne, Victoria, Australia

ABSTRACT

This study analysed the competition scheduling of future top 100 and 250 ranked tennis players from international tournament profiles at ages 13–18y. Retrospective tournament data were analysed for 165 future top 100 (T100) and top 250 (T250) males during their junior international tournament eligibility. Tournament/match volumes, days between tournaments and consecutive tournaments (<8 days between) were quantified for junior and professional events. A two-way (age x ranking) analysis of variance determined the effects of age and ranking group on tournament profiles. Significant interactions were observed for tournament volumes across junior and professional categories, with T100 players competing in professional tournaments earlier ($p < 0.05$). No significant interactions were observed for volumes of junior or professional matches played ($p > 0.05$). No significant interactions were observed for days between tournaments or consecutive tournaments played ($p > 0.05$). Significant main effects were observed for age on tournament volume, with junior and professional volume increasing at age 15 and 17, respectively ($p < 0.05$). Higher match volumes were observed for T100 players compared to T100-S players ($p < 0.05$). Competition schedules intensify at age 15 compared with ages 13–14y through increased tournament and match volumes. Future T100 players' transition to professional tournaments earlier, alongside greater engagement in higher quality junior tournaments.

ARTICLE HISTORY

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KEYWORDS

Racquet sports; junior development; athlete planning; player pathways

Introduction

Competition engagement through tournament play dictates the periodisation of tennis players' yearly calendars (Roetert et al., 2005). In turn, the accumulation of ranking points from tournaments provides the desired prestige and financial rewards for progression on the tour (Reid et al., 2014). This landscape is capricious due to the knockout structure of tournaments and irregularity of competition scheduling (Roetert et al., 2005). As a result, traditional periodisation strategies based on known or predictable durations of training and competition cycles are not easily designed or implemented in tennis (Roetert & Ellenbecker, 2009). Accordingly, many aspiring players prioritise competition over training, which can compromise the time available for dedicated sport-specific skill development that has been reported as being so central to the future success of elite athletes (Elferink-Gemser et al., 2011; Martens & Maes, 2005; Reid et al., 2009; Rumpf et al., 2014; Unierzyski, 2005). Tennis coaches working with talented junior players attempt to navigate these issues through strategic scheduling of competitions to balance the desire for short-term ranking improvements and long-term development (Unierzyski, 2003). Given the anecdotal nature of current literature on competition engagement, a data-driven understanding of typical competition engagement patterns of future successful players might

better inform training and athlete development recommendations.

Currently, most player development strategies in tennis define success at the individual player level as future attainment of a top 100 or top 250 professional ranking as these ranking milestones provide entry into Grand Slam events, which offer the game's most lucrative prize money (Bane et al., 2014; Brouwers et al., 2015; Reid & Morris, 2013). With that in mind, athlete selection into national development programmes is often based on a player's ranking status. Between the ages of 16–18 y, distinctive trajectories for future top 50 players exist, though rankings in early adulthood are more indicative of a player's future top 100 or top 250 status (Reid et al., 2014). Further, Li et al. (2020) have shown the poor predictive ability of professional ranking in adolescence to predict those players achieving a future ranking from 51 to 100. Ironically, as rankings are the product of competition results, it stands to reason that two players could achieve similar rankings in their developmental years through quite different engagements in competition scheduling and may represent a limitation of solely relying upon ranking for selection decisions. This understanding of the role of tournaments and therefore the organisation of aspiring professional players' competitive calendars is an obvious omission from previous work in tennis, and inadvertently contributes to the conjecture regarding the importance of performance at young ages (Brouwers et al., 2012; Unierzyski et al., 2003).

Information describing competition volumes and distributions to strategically plan players' schedules in tennis are limited to relatively crude accounts. Indeed, total international tournament volume at age 17 (11 ± 8 events) and 18 (15 ± 5 events) among future top 10 male players has been reported (Reid et al., 2009), with expert opinion pointing to lower volumes at 13–14 y as players prioritise sport-specific and physical skill development (Unierzyski, 2003, 2005). Coaching anecdotes suggest competition exposures in early adolescence should be managed so that players do not compete in any more than three consecutive tournaments (Unierzyski, 2005). Clearly, these types of time-honoured maxims are untested, meaning that guidelines for competition organisation/scheduling and long-term athlete development in tennis lack the necessary sophistication for effective implementation (Gerdin et al., 2020). Consequently, the aim of this study was to describe and compare the frequency and distribution of tournament-play across the junior pathway (13–18 y) of future top 100 and top 250 professionally ranked tennis players, while also outlining the practical implications for the modern coach and player.

Materials and methods

Participants

This retrospective study focused on international tournament engagement characteristics of future top 100 and 250 professionally ranked Association of Tennis Professionals (ATP) players between 13 and 18 years of age. Individual player data were analysed during their International Tennis Federation (ITF) junior tour eligibility (i.e., the day of 13th birthday to the end of 18th birth year) and further detailed according to the player's peak ATP ranking based on the available dataset. The initial sample of players obtained from the ITF included all players who competed in the main or qualifying draw at a junior or professional event from January 1st, 2000 through to December 31st, 2015. The earliest birth year possible for included players was the year 1987 to ensure that players' tournament activity between 13–18y could be captured. Player tournament and ranking data were obtained from publicly available domains including the official websites of the ITF and ATP. The ranking milestones for each player included the dates of their peak junior ranking, first professional ranking and entries into the top 100 (T100) and top 250 (T250). This study was approved by the University Human Research Ethics Committee (ETH19–3951).

Eligible players were those who achieved peak professional singles rankings inside the top 100 or from 101 to 250 during the aforementioned dates. With past research showing four years to be the average transition time from first professional ranking point to the top 100 (Kovacs et al., 2015; Reid & Morris, 2013), players were further categorised into the following groups for analysis;

- T100-fast (T100-F). T100 players achieving their ranking ≤ 4 years from first professional ranking point ($n = 42$),
- T100-slow (T100-S). T100 players achieving their ranking > 4 years from first professional ranking point ($n = 55$),

- T250 (T250). Players achieving a T250 rank and meeting at least one of the following criteria ($n = 68$):
 - Are ≤ 8 y removed from first professional ranking and have been in T250 for > 4 y
 - Are > 8 y removed from first professional ranking and have been in T250 for > 4 y
 - Are > 8 y removed from first professional ranking and have been in T250 for ≤ 4 y

Data collation

Junior and professional international tournaments were categorised according to the possible ranking points earned and thus used to represent tournament quality. For the junior tour, Grade A and Grade 1 junior ITF events represented Category 1 tournaments, Category 2 tournaments included Grade 2 and Grade 3 junior ITF events, while Category 3 included Grade 4 and Grade 5 junior ITF events. For the professional tour, Category 1 tournaments were the four Grand Slams, Category 2 were ATP World Tour tournaments, Category 3 were ATP Challenger events and Category 4 included ITF Futures/Satellites. Local tournaments administered by the respective national tennis federations were not considered.

Annual competition engagement was described through total matches played, total junior matches played, total professional matches played, days between tournaments and number of consecutive tournaments. For total matches only, further analysis was undertaken which detailed total matches played per month based on a count per calendar month. Consecutive tournaments were defined as any tournament, regardless of tour or category, that started < 8 days from the player's previous tournament ending. This definition was determined from the reporting of dates in the dataset. Walkovers were noted and excluded from the match analysis. Matches played were inclusive of both singles and doubles matches. To account for instances where eligible players did not play tournaments in a given year, tournament and match volumes were reflected as '0'.

Statistical analysis

All statistical analysis was performed in the R language (RStudio, 1.1.463, RStudio, Inc.). The mean and standard deviation were reported for all tournament and match variables and reported annually, and for each birth year. Data normality was assessed via a Shapiro-Wilk test and revealed all variables were not normally distributed. Data were then log-transformed prior to analysis. A two-way (age x ranking group) analysis of variance (ANOVA) determined the effects of respective age and ranking groups on competition engagement metrics. Tukey's post-hoc test was used with a Bonferroni correction to reduce risk of Type I error. Significance was set at 0.05.

Results

Annual junior tournaments played

Figure 1 shows the annual junior tournaments played across ages for the three pre-defined junior tournament categories.

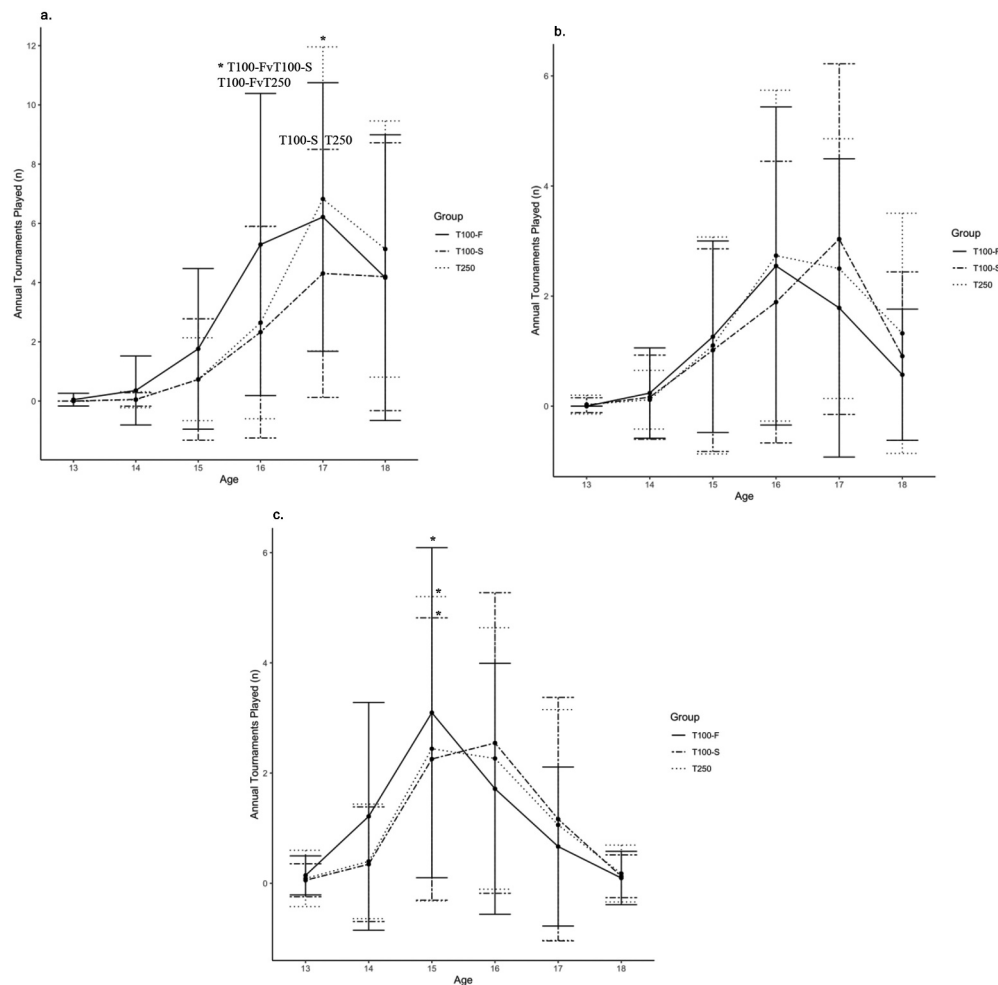


Figure 1. Annual Junior International Tennis Federation (ITF) tournaments played. All data presented as mean $\pm$ standard deviation (a) Junior Category 1 (Junior ITF Grade A and Grade 1). (b) Junior Category 2 (Junior ITF Grade 2 and Grade 3). (c) Junior Category 3 (Junior ITF Grade 4 and Grade 5). **Groups** T100-F. Players who made the top 100 ≤ 4 years of first professional ranking point T100-S. Players who made the top 100 more than >4 years after first professional ranking point T250. Players who achieved a ranking inside 101–250* significantly different from previous age for same ranking group ($p < 0.05$). Differences between ranking group for same age are denoted (example T100-FvT250) ($p < 0.05$).

Significant interaction effects were observed for age and ranking group in junior category 1 ($p < 0.01$; **Figure 1A**). Significant main effects were observed for age ($p < 0.01$), with post-hoc analyses showing significant increases in tournaments at age 16 and 17 ($p < 0.01$), followed by a trend towards reduced tournament volume at age 18 ($p = 0.06$). A significant main effect was observed for ranking group in junior category 1 tournaments played ($p < 0.01$). Category 1 tournaments played by T100-F players were significantly greater than T100-S players ($p < 0.01$).

For junior tournaments in category 2, no significant interaction effects were observed ($p = 0.06$). Significant main effects were observed for age ($p < 0.01$), whereby ages 15 and 16 showed a significant increase in junior category 2 tournaments played ($p < 0.01$). A significant reduction in tournaments played was observed at age 18 compared to age 17 ($p < 0.01$). No significant main effects were observed for ranking group ($p = 0.17$).

Significant interaction effects were observed for age and ranking group for category 3 junior tournaments ($p = 0.04$;

Figure 1C). Significant main effects were found for age ($p < 0.01$); with post-hoc tests revealing a significant increase in tournaments played at age 15 from earlier ages ($p < 0.01$), with an ensuing decrease in tournaments played at age 17 and 18 ($p < 0.01$ and $p < 0.01$, respectively). No significant main effects were observed for ranking group on tournaments played in this category ($p = 0.47$).

Annual professional tournaments played

Figure 2 shows the annual tournaments played across ages for all four professional tournament levels. For Grand Slam competitions (professional category 1), significant interaction effects were observed for age and ranking group ($p < 0.01$; **Figure 2A**). Significant main effects for age were evident ($p < 0.01$), with a significant increase in Grand Slams played at age 18 compared with previous ages ($p < 0.01$). Significant main effects were observed for ranking group ($p < 0.01$), with an increase in category 1 tournaments for T100-F players compared with T100-S and T250 players ($p < 0.01$).

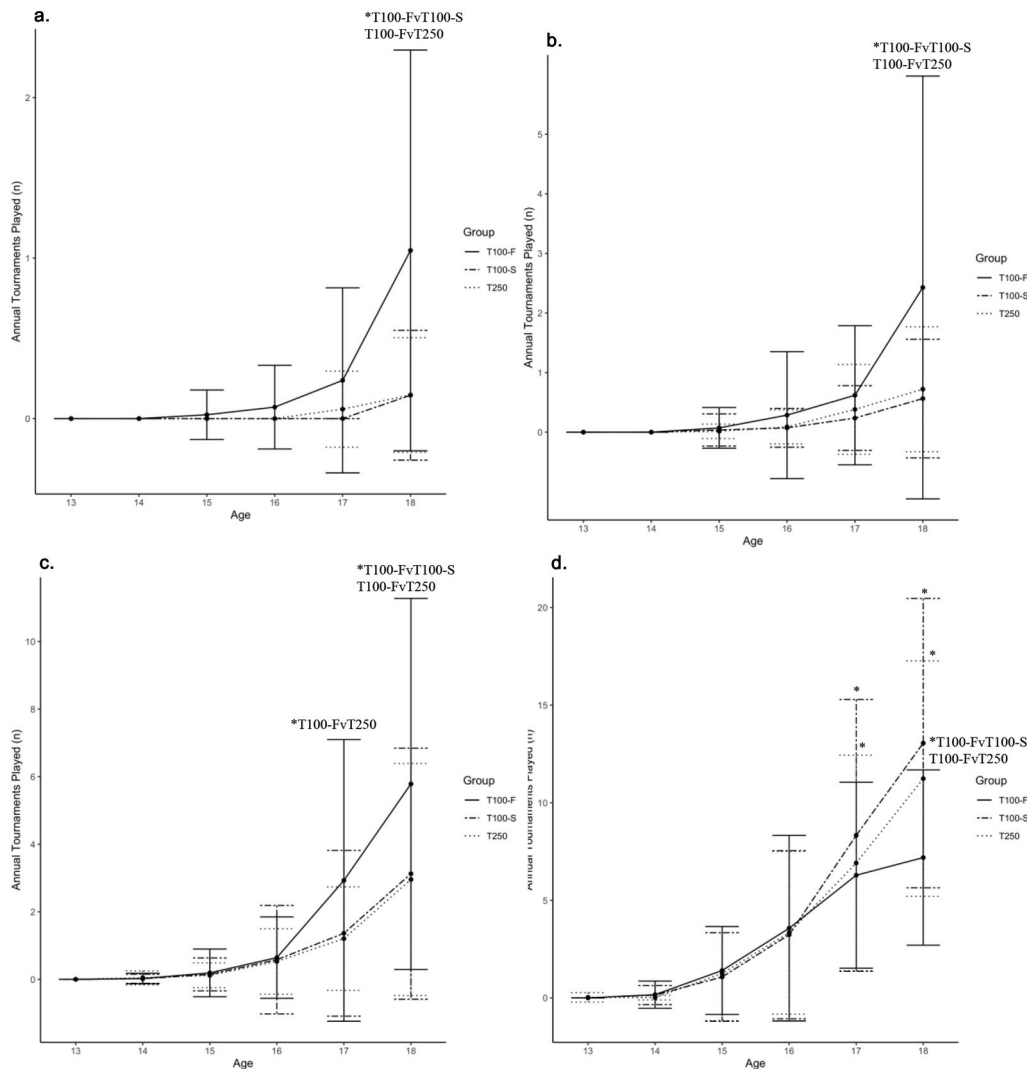


Figure 2. Annual Association of Tennis Professionals (ATP) and Men's Tour International Tennis Federation (ITF) tournaments played. All data presented as mean $\pm$ standard deviation. (a) Professional Category 1 (Grand Slams). (b) Professional Category 2 (ATP World Tour Tournaments). (c) Professional Category 3 (ATP Challengers). (d) Professional Category 4 (ITF Futures/Satellites). **Groups** T100-F. Players who made the top 100 ≤ 4 years of first professional ranking point T100-S. Players who made the top 100 more than >4 years after first professional ranking point T250. Players who achieved a ranking inside 101–250* significantly different from previous age for same ranking group ($p < 0.05$). Differences between ranking group for same age are denoted (example T100-FvT250) ($p < 0.05$).

Figure 2B shows significant interaction effects observed for age and group on tournaments played in category 2 ($p < 0.01$). Significant main effects existed for age ($p < 0.01$), with post-hoc analyses revealing an increase at age 18 ($p < 0.01$). Additionally, significant main effects were observed for ranking group ($p < 0.01$), where category 2 tournaments were played more by those in T100-F ($p < 0.01$).

Significant interaction effects for age and group on tournaments played in professional category 3 were evident ($p < 0.01$; Figure 2C). Significant main effects for age revealed an increase in tournaments played at ages 17 and 18, compared to younger ages ($p < 0.01$). For ranking group, significant main effects were also observed ($p < 0.01$), with tournament volume by T100-F players significantly higher ($p < 0.01$).

Significant interaction effects for age and group were found across professional tournaments played in category 4 ($p < 0.01$; Figure 2D). Significant main effects were found for age ($p < 0.01$) and revealed a significant increase in tournaments played at age 16, 17 and 18 ($p < 0.01$). Significant main effects

were found for ranking groups on category 4 tournaments played ($p < 0.01$) and, post-hoc analyses, revealed an increase in tournaments for T100-F players compared with T100-S ($p < 0.01$).

Annual match and tournament distribution variables

There were no significant interaction effects for the total volume of matches played ($p = 0.50$; Table 1). Significant main effects were observed for age ($p < 0.01$), with increases in matches played at ages 16 and 17 ($p < 0.01$). Significant main effects were observed for ranking group ($p < 0.01$), with total matches significantly greater in T100-F compared to T100-S ($p < 0.05$). Figure 3 shows the distribution of matches played per month for players grouped via future T100 and T250 status across ages 13 to 18 y. The figure depicts an apparent peak increase in density of matches played per month occurring at ages 16–18 y.

No significant interaction effects were observed for age and group on junior matches played ($p = 0.44$). Main effects were

Table 1. Annual matches played and tournament distribution variables.

A.				B.			
Group				Group			
Age	T100-F^(T100-S)	T100-S	T250	Age	T100-F	T100-S	T250
13	0.50 ± 1.33	0.24 ± 1.10	0.35 ± 1.90	13	0.50 ± 1.33	0.24 ± 1.10	0.32 ± 1.89
14	7.81 ± 14.95	2.25 ± 7.45	2.07 ± 7.72	14	7.50 ± 14.92	1.96 ± 7.35	2.01 ± 7.73
15	31.36 ± 28.51	19.02 ± 24.03	20.25 ± 25.11	15	28.29 ± 26.07	16.45 ± 22.46	17.40 ± 22.34
16*	61.26 ± 37.24	40.75 ± 31.14	44.21 ± 31.92	16*	47.26 ± 31.02	31.31 ± 27.23	34.37 ± 25.62
17*	80.88 ± 31.44	71.22 ± 35.31	74.53 ± 32.77	17	47.24 ± 31.18	41.73 ± 32.83	49.60 ± 30.95
18	83.83 ± 24.02	84.76 ± 33.40	84.56 ± 27.00	18*	27.43 ± 29.42	27.40 ± 31.17	34.25 ± 27.27
C.				D.			
Group				Group			
Age	T100-F	T100-S	T250	Age	T100-F	T100-S	T250
13	0.00 ± 0.00	0.00 ± 0.00	0.03 ± 0.24	13	22.00 ± 0.00	7.00 ± 0.00	64.60 ± 59.61
14	0.31 ± 1.33	0.29 ± 1.08	0.06 ± 0.24	14	51.11 ± 85.71	64.80 ± 94.18	71.36 ± 88.92
15	3.07 ± 5.77	2.56 ± 5.29	2.85 ± 6.57	15*	33.40 ± 64.16	34.72 ± 58.31	40.14 ± 69.54
16	14.00 ± 20.35	9.44 ± 14.80	9.84 ± 13.82	16	24.39 ± 35.61	31.62 ± 48.73	28.75 ± 48.32
17*	33.64 ± 31.03	29.49 ± 26.51	24.93 ± 20.22	17*	19.85 ± 28.04	19.86 ± 30.78	19.07 ± 27.17
18*	56.40 ± 27.00	57.36 ± 34.32	50.31 ± 25.78	18	16.49 ± 21.57	15.93 ± 22.33	16.86 ± 22.25

All data presented as mean ± standard deviation.

(A)

(A) Annual Professional and Junior Tour Matches Played. (B) Annual Junior Tour Matches Played. (C) Annual Professional Tour Matches Played (D) Days Between Tournaments.

(A) Annual Professional and Junior Tour Matches Played. (B) Annual Junior Tour Matches Played. (C) Annual Professional Tour Matches Played (D) Days Between Tournaments.

Groups

T100-F. Players who made the top 100 ≤ 4 years of first professional ranking point.

T100-S. Players who made the top 100 more than >4 years after first professional ranking point.

T250. Players who achieved a ranking inside 101–250.

* significantly different from previous age ($p < 0.05$).

(T100-S) significantly different from T100-S ($p < 0.05$).

(T250) significantly different from T250 ($p < 0.05$).

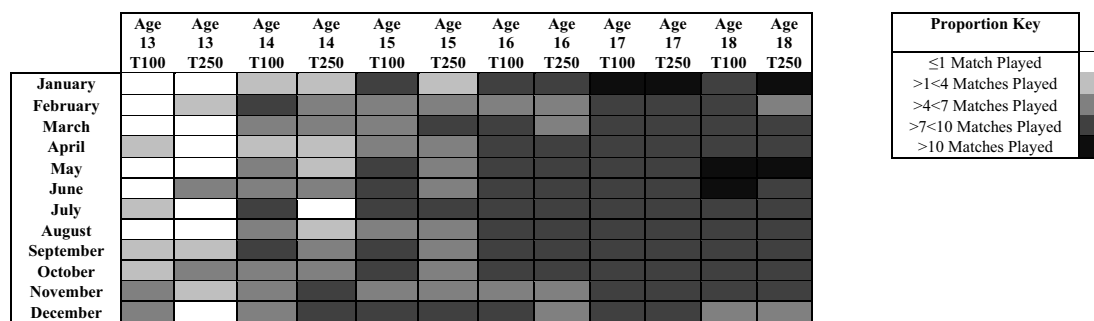


Figure 3. Average matches played per month for future top 100 (T100) and 250 (T250) players.

observed for age ($p < 0.01$), and post-hoc analyses revealed that junior matches increased at ages 16 ($p < 0.01$) with a subsequent reduction at age 18 ($p < 0.01$). No significant main effects were observed for ranking group ($p > 0.05$).

No significant interaction effects were observed ($p = 0.89$) for professional matches played. Significant main effects were found for age ($p < 0.01$), with professional matches played increasing at ages 17 and 18 ($p < 0.01$). No significant main effects for group were observed ($p = 0.08$).

For days between tournaments, no significant interaction effects were observed for age and ranking group ($p = 0.85$). Significant main effects were observed for age ($p < 0.01$), with post-hoc tests showing increased tournament density at ages 15 and 17 compared to other ages ($p < 0.01$). No significant main effects were observed for ranking group ($p = 0.15$).

For consecutive tournaments played, no significant interaction effects for age and group were observed ($p = 0.67$; Table 2). Significant main effects for age were found ($p < 0.01$) with post-hoc tests revealing an increase in consecutive tournaments played at age 16, 17 and 18 ($p < 0.01$). No significant main effect was observed for ranking group ($p = 0.15$).

Discussion

The aim of this study was to describe and compare the frequency and distribution of tournament-play across the junior pathway (13–18 y) of future top 100 and top 250 professionally ranked tennis players. Tournament volumes increased from age

15, with late adolescence characterised by an increased number of professional tournaments played by future T100 players. Whilst match volumes progressively increased from age 16, the lack of interaction effect suggests similar overall match-play engagement. This is instructive as it highlights that the degree of future success is not simply a function of having greater access to international competition volumes when young; a point previously made by Brouwers et al. (2012), who highlighted that competitive junior performances had only limited effect on a player's chances of future professional success. The tournament profiles of T100-F players did reveal however that earlier engagement in higher quality tournaments may be associated with a greater degree of future success, supporting previous suggestions (Li, De Bosscher, Weissensteiner et al., 2018a; Li et al., 2020). These insights of the competition scheduling patterns of successful professional players are needed to support the previously anecdotal long-term athlete development (LTAD) recommendations in tennis. Whilst it is acknowledged that development pathways of individual athletes are unique, these findings appear to show general competition engagement metrics for players to strive, and national federations to steer recommendations, towards.

The 18 and under competition profiles appear to illustrate how priorities and foci change throughout adolescence. At ages 13–14 y, it has previously been suggested that tournament exposures are tightly controlled (i.e., ≤ 9 ITF tournaments annually) to allow optimal time for development of skill and physical capacities (Unierzyski, 2003, 2005). This appears to be true of the players in this sample as junior tournament volumes were comparable between ages 13–14 and across ranking categories. This reduced international tournament play would infer players at this age dedicate greater time to training and sport-specific development, as has been observed in other sports (Huijgen et al., 2013). Indeed, capitalising on the training opportunities in early adolescence are critical in accumulating the sport-specific training needed to develop expertise and enhance chances of success in later adolescence (Monsaas, 1985), thus the observed lower emphasis on international competitions. Furthermore, this accumulation of training time in early adolescence may assist players in developing the skills necessary to facilitate their transition to senior competitions, which was associated with future success in combat sport (i.e., taekwondo, wrestling and boxing) athletes (Li, De Bosscher, Pion et al., 2018b). Alternatively, it could be that early adolescence typically involves the concurrent sampling of multiple

Table 2. Annual consecutive[^] tournaments played.

Age	Group		
	T100-F	T100-S	T250
13	0.00 ± 0.00	0.33 ± 0.58	0.20 ± 0.45
14	1.33 ± 1.46	0.85 ± 1.21	0.33 ± 0.91
15	3.14 ± 3.06	2.72 ± 2.75	2.52 ± 3.26
16*	6.20 ± 5.24	4.37 ± 4.37	5.42 ± 4.60
17*	9.12 ± 5.60	9.35 ± 5.66	9.47 ± 4.97
18*	11.07 ± 4.09	12.55 ± 6.31	11.74 ± 4.84

All data presented as mean ± standard deviation.

[^]Consecutive tournament defined as those occurring less than 8 days apart of each other.

Groups

T100-F. Players who made the top 100 ≤ 4 years of first professional ranking point.

T100-S. Players who made the top 100 more than >4 years after first professional ranking point

T250. Players who achieved a ranking inside 101–250.

* significantly different from previous age ($p < 0.05$).

sports (Baker et al., 2003; Cote, 1999), which might also account for the lower tournament volumes observed during this development period. More realistically though, historical perspectives in tennis suggest that players will have sourced domestic or regional events (not analysed in this study) to complement their international exposure (Monsaas, 1985) and that these lighter international schedules may even be a function of the sport's international age eligibility restrictions (ITF, 2016).

With international tournament volume clearly increasing from age 15 onwards in future T100 and T250 players, it suggests that players at these ages begin to prioritise international tennis competition over other commitments. Using T100 players in their 16th BY as an example, ~14 weeks of the year involve international competition. Current recommendations from Tennis Australia suggest 15–17y players compete in relatively conservative 8–17 international tournaments per annum. Increasing tournament volumes impacts the available training time during a key development period and previous analysis of elite Australian athletes has revealed that the average individual sport athlete enters the highest level of junior competition at age 16 ± 2 y and experience a ≈ 4 h increase in training duration per week (Gulbin et al., 2010). It is unclear whether similar increases in training volume are evident in tennis players and represents a limitation of the current dataset in that the training profiles of players were not captured. Inclusions of such profiles would provide a deeper understanding of competition and training engagement to inform LTAD guidelines in tennis. Further benefits of such data relate to the appropriate timing of training stimuli (physical and skill) around such condensed periods of competition.

Experts opine that a maximum of three consecutive tournaments should feature in annual plans (Unierzyski, 2005), yet our results show that ≈ 5 –10 consecutive tournaments are regularly played throughout late adolescence. This amalgam of consecutive tournaments likely compromises a player's physical conditioning due to a lack of appropriate training stimulus (Murphy et al., 2015) and clearly presents as a programmatic challenge for players and coaches. Outlined visually in Figures 3, 7–9 matches per month are consistently played throughout the year from age 16 and, combined with the nature of consecutive tournaments, further challenges the planning processes of physical preparation staff. This may be necessary though as, given the future success achieved by players in the current sample, it is possible that the increased density of monthly matches results from reaching the latter rounds of tournaments, though this remains speculative.

Longitudinal accounts of match volumes have not been reported in the literature; however, leaders in Belgian tennis have documented that top 100 junior ranked girls (16.2 y) played between 43 and 74 matches per annum with a yearly average of 55 matches (Martens & Maes, 2005). The current study found similar average annual match volumes played at age 16, with a range of 44–61/year. Indeed, match-play volumes followed a similar pattern to overall tournament engagement, with significant increases noted at age 15 through to 17 y. This is to be expected given guidelines from federations suggest stepped increases in match-play

volume from age 15+, culminating in 80–100 singles matches at the age of 17y. Whilst the average matches at ages 17 and 18 in the present study are towards the lower end of these recommendations, it is likely a result of local domestic competitions not being captured. These final stages of adolescence were further characterised by increasing engagement on the professional circuits, which is consistent with what is commonly reported.

Previous surveys of elite athlete pathways have highlighted the practical relevance of competition milestones, with future elite-level athletes entering professional competitions as teenagers (Gulbin et al., 2010). Further emphasising this point, Li, De Bosscher, Pion et al. (2018b) reported higher success rates of future senior combat sport athletes (in particular taekwondo) when succeeding at senior events in their junior years. Our results appear similar, where future T100 players competed in a higher number of ATP Tour events at ages 17 and 18, and likely influenced their eventual professional status. This is to be expected given the previously observed distinctiveness of ranking trajectories for future top 50 players at ages 16–18 (Reid et al., 2014) likely resulting from an earlier engagement in ATP-level events (Li, De Bosscher, Weissensteiner et al., 2018a), and appears to highlight the emergence of precocious talent in tennis. Although future 51–100 and 101–175 ranked players did not have distinctive ranking signatures at these ages (Li et al., 2020; Reid et al., 2014), the present study showed T100 players did engage earlier in more professional competitions compared to future T250 players in late adolescence. This suggests that current selection or LTAD guidelines, anchored alone in adolescent ranking data, may oversimplify the current performance and future ranking trajectory of players. This may also suggest future selection guidelines aim to consider the quality of tournaments played leading to a player's resultant ranking status.

Limitations

Whilst the strength of this study is the novel reporting of competition engagement in future successful tennis players, limitations exist in the metrics reported. First, the inclusion of doubles matches may be seen as a limitation of true ranking progression in singles; however, was reported in order to provide an overall understanding of competitive load. This study also reports exclusively on international competitions without consideration to domestically sanctioned tournaments. Thus, it is likely that the tournament and match volumes are underestimated for players, especially those under the age of 17 where age eligibility limits international tournament engagement (ITF, 2016). Similarly, the assumption that when players were not competing in these tournaments, they were in periods of focused training or recovery is likely imperfect. Maturation status of players was also not considered in this regard to provide greater context to participation, or lack thereof, during adolescence. This study does also not account for other reasons for non-participation in tournaments at a given age (i.e., injury) and, as a result, is only able to provide limited context to explain the speed of a player's transition to the T100. Further, the groupings of players may have masked more subtle differences among

smaller groups of ranked players (i.e., top 10 players versus top 50–100 players). With regard to classification of tournament qualities, it is acknowledged that the groupings used are imperfect given changes in ranking point allocations over the years in question. Lastly, it should be noted that the introduction of the ITF World Tour in 2019 changed the landscape of entry-level professional events and may impact the generalisability of the present findings.

Conclusion

This study has outlined the international tournament scheduling characteristics of future T100 and T250 male players throughout their junior tournament eligibility (13–18 y). For aspiring male players, international competition schedules intensify in their volume and distribution from age 15. However, whilst this intensification is consistent among future T100 and T250 players, the quality of tournaments played throughout this period appears to be a key factor in distinguishing the degrees of success achieved. The present study suggests that coaches and national federations are aware of the intensification of international tournament volumes at age 15 and that recommendations from federations in the formative adolescent years provide players with guidelines to manage this increased competitive load. Given the competing interests that appear to exist in order to maximise competitive opportunities whilst continuing to develop sport-specific skills and physical capacities, further research is needed to analyse the training patterns of future successful players throughout this critical period of development.

Disclosure statement

No potential conflict of interest was reported by the authors.

ORCID

Rob Duffield  <http://orcid.org/0000-0002-5641-1314>

References

- Baker, J., Horton, S., Robertson-Wilson, J., & Wall, M. (2003). Nurturing sport expertise: Factors influencing the development of elite athlete. *Journal of Sports Science & Medicine*, 2(1), 1–9. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3937568/>
- Bane, M. K., Reid, M., & Morgan, S. (2014). Has player development in men's tennis really changed? An historical rankings perspective. *Journal of Sports Sciences*, 32(15), 1477–1484. <https://doi.org/10.1080/02640414.2014.899706>
- Brouwers, J., De Bosscher, V., & Sotiropoulos, P. (2012). An examination of the importance of performances in youth and junior competition as an indicator of later success in tennis. *Sport Management Review*, 15(4), 461–475. <https://doi.org/10.1016/j.smr.2012.05.002>
- Brouwers, J., Sotiropoulos, P., & De Bosscher, V. (2015). An examination of the stakeholders and elite athlete development pathways in tennis. *European Sport Management Quarterly*, 15(4), 454–477. <https://doi.org/10.1080/16184742.2015.1067239>
- Cote, J. (1999). The influence of the family in the development of talent in sport. *The Sport Psychologist*, 13(4), 395–417. <https://doi.org/10.1123/tsp.13.4.395>
- Elferink-Gemser, M. T., Jordet, G., Coelho-E-Silva, M. J., & Visscher, C. (2011). The marvels of elite sports: How to get there? *British Journal of Sports Medicine*, 45(9), 683–684. <https://doi.org/10.1136/bjsports-2011-090254>
- Gerdin, G., Fahlstrom, P. G., Glemne, M., & Linner, S. (2020). Swedish tennis coaches' everyday practices for creating athlete development environments. *International Journal of Environmental Research and Public Health*, 17(12), 4580. <https://doi.org/10.3390/ijerph17124580>
- Gulbin, J. P., Oldenziel, K., Weissensteiner, J. R., & Gagne, F. (2010). A look through the rear view mirror: Developmental experiences and insights of high performance athletes. *Talent Development & Excellence*, 2(2), 149–164.
- Huijgen, B. C. H., Elferink-Gemser, M. T., Ali, A., & Visscher, C. (2013). Soccer skill development in talented players. *International Journal of Sports Medicine*, 34(8), 720–726. <https://doi.org/10.1055/s-0032-1323781>
- ITF 2016, ITF Rules of tennis. International Tennis Federation. <<http://www.itf.tennis.com/about/organisation/rules.aspx>>.
- Kovacs, M., Mundie, E., Eng, D., Bramblett, J., Kovacs, M. J., & Hosek, R. (2015). How did the top 100 professional tennis players (ATP) succeed: An analysis of ranking milestones. *Journal of Medicine and Science in Tennis*, 20(2), 50–57.
- Li, P., De Bosscher, V., Pion, J., Weissensteiner, J. R., & Vertonghen, J. (2018b). Is international junior success a reliable predictor for international senior success in elite combat sports? *European Journal of Sport Science*, 18(4), 550–559. <https://doi.org/10.1080/17461391.2018.1439104>
- Li, P., De Bosscher, V., & Weissensteiner, J. R. (2018a). The journey to elite success: A thirty-year longitudinal study of the career trajectories of top professional tennis players. *International Journal of Performance Analysis in Sport*, 18(6), 961–972. <https://doi.org/10.1080/24748668.2018.1534197>
- Li, P., Weissensteiner, J. R., Pion, J., & De Bosscher, V. (2020). Predicting elite success: Evidence comparing the career pathways of top 10 to 300 professional tennis players. *International Journal of Sports Science & Coaching*, 15(5–6), 793–802. <https://doi.org/10.1177/1747954120935828>
- Martens, S., & Maes, ca. 2005, 'Periodisation for professional female tennis players', *ITF Coaching and Sport Science Review*, no. 36.
- Monsaas, J. A. (1985). Learning to be a world class tennis player. In B. S. Bloom (Ed.), *Developing talent in young people*, 211–269. Ballantine Books.
- Murphy, A. P., Duffield, R., Kellett, A., & Reid, M. (2015). The relationship of training load to physical-capacity changes during international tours in high-performance junior tennis players. *International Journal of Sports Physiology and Performance*, 10(2), 253–260. <https://doi.org/10.1123/ijsp.2014-0038>
- Reid, M., Morgan, S., Churchill, T., & Bane, M. K. (2014). Rankings in professional men's tennis: A rich but underutilized source of information. *Journal of Sports Sciences*, 32(10), 986–992. <https://doi.org/10.1080/02640414.2013.876086>
- Reid, M., & Morris, C. (2013). Ranking benchmarks of top 100 players in men's professional tennis. *European Journal of Sport Science*, 13(4), 350–355. <https://doi.org/10.1080/17461391.2011.608812>
- Reid, M., Quinlan, G., Kearney, S., & Jones, D. (2009). Planning and periodization for the elite junior tennis player. *Strength and Conditioning Journal*, 31(4), 69–76. <https://doi.org/10.1519/SSC.0b013e3181afc98d>
- Roetert, E., & Ellenbecker, T. (2009). Periodization training. *ITF Coaching and Sport Science Review*, 16(47), 10–11.
- Roetert, E. P., Reid, M., & Crespo, M. 2005, 'Introduction to modern tennis periodisation', *ITF Coaching and Sport Science Review*, no. 36, p. 2.
- Rumpf, M. C., Schneider, A. S., Schneider, C., & Mayer, H. M. (2014). Training profiles and motivation of male and female youth soccer players. *International Journal of Sports Science & Coaching*, 9(1), 207–216. <https://doi.org/10.1260/1747-9541.9.1.207>
- Unierzyski, P. 2003, 'Planning and periodisation for the 12–14 year old tennis players', *ITF Coaching and Sport Science Review*, no. 31, pp. 6–7.
- Unierzyski, P., Wielinski, D., & Zhanel, J. (2003). Searching reasons of success and failure of careers of young tennis players - study of two individual cases. In S. Miller (Ed.), *Tennis Science and Technology*. 253–259. International Tennis Federation.
- Unierzyski, P. 2005, 'Periodisation for under-14s', *ITF Coaching and Sport Science Review*, no. 36, pp. 4–6.